# Range

**What it is.** The range is the difference between the highest and lowest value.

**Important.** When the range is big, it means that there is large variety in the dataset. If the range is small, it means that there is little variety in the dataset.

# Mean

**What it is.** The value obtained by adding all values of a set number of observations divided by the number of observations. The mean is also known as the average.

**Important.** The mean is often used as a measure for central tendency`in symmetric distributions. Since outliers can have a large effect on the mean, for skew distributions, we prefer the median as a measure of location.

## Sensitivity

**What it is.** Sensitivity (also called the true positive rate) measures the proportion of actual positives that are correctly identified as such (e.g., the percentage of diseased people who are correctly identified as having the condition). (30)

**Important.** When the sensitivity is high, it means that a low amount of people who are actually ill are tested negative on that test. When the sensitivity is low it means that a lot of people who are ill are missed.

## Variable

**What it is.** A characteristic or numerical value which can take on different values.

**Important.** nan

## Population

**What it is.** The population is a well defined set of subjects or objects A statistical population can be a group of existing objects. This group can potentially be of infinite size.

**Important.** nan

## Event

**What it is.** A thing that happens or takes place.

**Important.** nan

## Sample

**What it is.** A proportion of subjects collected from a population on which you are going to perform research.

**Important.** nan

## Unit

**What it is.** 1. Unit of observation: the who or what within a particular research on which data is collected 2. Unit of measurement: the type of measurement in which a certain magnitude is expressed

**Important.** nan

for **Variable** see page 2

for **Sample** see page 2

for **Mean** see page 1

for **Population** see page 2

# Outcome

**What it is.** The outcome is the result of a trial of a probability experiment. Outcome is also used for “outcome variable” or “response variable”.

**Important.** The outcome is the end result of a conducted research. It usually is a list of statistics / probabilities or it could be a conclusion.

# Demographics

**What it is.** Demographics is the study of the structure of populations. Demographics is useful for learning about certain characteristics of a population.

**Important.** nan

for **Event** see page 2

for **Mean** see page 1

for **Population** see page 2

# Median

**What it is.** The median is the middle variable in a number of ordered observations. If the dataset has an even number, it is calculated by the taking the mean of the two observations in the middle. The median is used as a measure of location in distributions of continuous and ordinal variables.

**Important.** The median is often used as a measure for central tendency.

for **Range** see page 1

# Box Plot

**What it is.** In descriptive statistics, a box plot is a method of graphically illustrating how the data within the data set is distributed.

**Important.** The box represents the middle 50% of the data (interquartile range). The line which divides the box into two parts is the median (most middle value within the data set). Box plots may also have lines extending outside of the box (whiskers). These whiskers represent the total range of the data by indicating the upper and lower boundaries of the data set. There may also be outliers/extremes which are shown by dots either below or above the whisker lines.

for **Outcome** see page 3

# Exposure

**What it is.** Exposure refers to all external factors which influence or affect something

**Important.** Exposure are all external factors and should be accounted for when conducting analysis, since exposure can influence the testing variable. If it is not accounted for, it can skew the result positively or negatively.